

Week 1: Circuit Concepts

Charge, current, voltage, power, KCL, KVL, series/parallel, dividers, equivalent resistance

In-depth notes covering all core material of this chapter: theory, engineering shortcuts, exam traps, 17 worked examples, and 8 practice problems with full solutions.

Based on: Rizzoni & Kearns, *Principles and Applications of Electrical Engineering*, 7th ed., §1.1–1.6 and §2.1–2.3

Inspired by: ELEC 275, Principles of Electrical Engineering, Concordia University

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How to use this deck

- **“Try it” slides:** the problem alone, with empty space. Do it first.
- **Next slide:** the clean solution, step by step.
- **Purple boxes:** engineering shortcuts, the fast way to see or solve things.
- **Orange boxes:** traps. This is where marks get lost on exams.
- Formulas are tagged **[on formula sheet]** or **[NOT on formula sheet]**, based on the sheet given in ELEC 275 exams.
- **End of deck:** 8 final-check problems, then a full answer sheet.

The right side of each slide is left empty for your notes.

Symbols and units

Symbol	Name	Unit	Plain meaning
q	charge	coulomb (C)	amount of electricity
i, I	current	ampere (A)	charge flowing past a point per second
v, V	voltage	volt (V)	energy per charge <i>between two points</i>
R	resistance	ohm (Ω)	how much it blocks current
G	conductance	siemens (S)	$1/R$, how easily current flows
p, P	power	watt (W)	energy per second
w, W	energy	joule (J)	total work done

Lowercase v, i = can change with time. Uppercase V, I = constant (DC). Weeks 1–3 are all DC.

Charge, current, voltage, energy

[NOT on formula sheet: memorize]

Current $i = dq/dt$ constant current: $q = I t$

Voltage $v = dw/dq$ energy per unit charge

Power $p = dw/dt = v i$

Energy $w = \int p dt$ constant power: $W = P t$

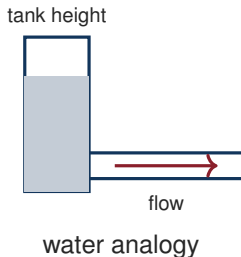
1 A = 1 C/s 1 V = 1 J/C 1 W = 1 J/s

1 A·h = 3600 C 1 kWh = 3.6 MJ

Trap

A·h (amp-hour) is **charge**, not energy. Energy = voltage $\times$ charge.

What these quantities mean



Charge q : the “amount of electricity” (like litres of water).

Current $i = dq/dt$: how fast charge moves past a point (litres per second).

Voltage $v = dw/dq$: energy each coulomb gains or loses between two points (the height difference the water falls).

Power $p = vi$: energy per second. Height $\times$ flow: more drop or more flow means more power.

Energy w : power added up over time (joules).
“ dw ” means a small bit of energy.

Note: voltage is always **between two points**. “The voltage at a node” means measured from the reference node (ground).

Units trick: $\text{k}\Omega \times \text{mA} = \text{V}$

Engineering shortcut: kilo-ohms and milliamps cancel

$$\text{k}\Omega \times \text{mA} = \text{V}$$

$$\text{V} \div \text{k}\Omega = \text{mA}$$

Example: $5 \text{ V} / 2 \text{ k}\Omega = 2.5 \text{ mA}$. No 10^3 or 10^{-3} needed.

$$\text{m} = 10^{-3} \quad \text{k} = 10^3 \quad \mu = 10^{-6} \quad \text{M} = 10^6$$

Trap

$\text{mA} \times \Omega$ gives **mV**, not V. Exams often mix Ω and $\text{k}\Omega$ in one circuit, so convert everything to the same unit first.

Try it: Example 1, units

Example 1

- (a) 12 V across 4 k Ω . Current?
- (b) 2.5 mA through 2 k Ω . Voltage?
- (c) A 12 V battery is rated 60 A·h. How much charge (C) and energy (J and Wh) does it hold?
- (d) A 100 W lamp runs for 5 h. Energy in kWh and in J?

your work here ↓

Example 1: solution (1/2)

(a) Ohm's law $I = V/R$, and $V \div k\Omega$ gives mA: $12/4 =$

3 mA

(b) Ohm's law $V = IR$, and $\text{mA} \times k\Omega$ gives V: $2.5 \times 2 =$

5 V

(c) charge 1 A·h = 1 A for 3600 s: $q = It = 60 \times 3600 =$

216 000 C

Example 1: solution (2/2)

(c) **energy** energy = voltage $\times$ charge: $W = Vq = 12 \times 216\,000 =$
2.59 MJ

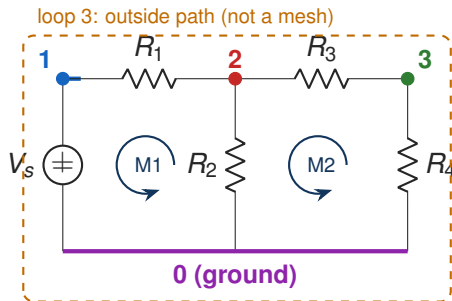
same in Wh: $12\text{ V} \times 60\text{ A}\cdot\text{h} =$ **720 Wh** ($720 \times 3600\text{ J}$
 $= 2.59\text{ MJ}$ ✓)

(d) constant power: $W = Pt = 100\text{ W} \times 5\text{ h} = 500\text{ Wh} =$
0.5 kWh

$1\text{ kWh} = 3.6\text{ MJ}$: $0.5 \times 3.6 =$ **1.8 MJ**

Shortcut: Wh $\rightarrow$ J and A·h $\rightarrow$ C both multiply by 3600 (seconds in an hour).

Nodes, branches, loops, meshes



Branch: one element. Here: 5

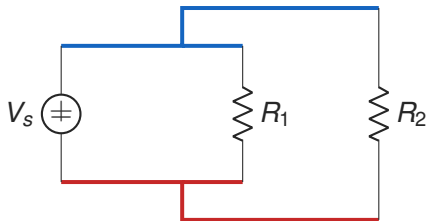
Node: a connection point (colored dots). Here: 4

Loop: any closed path. Here: 3

Mesh: a loop with nothing inside it, like a window. Here: M1, M2

Loop 3 goes around the outside through V_s , R_1 , R_3 , R_4 . It contains M1 and M2, so it's **not** a mesh.

Most important rule: one wire = one node



All blue wire is **one node**. All red wire is **one node**.

Only **2 nodes**, so V_s , R_1 , R_2 all have the same voltage.

Wire length and shape never matter.

Ideal wire and ground

Ideal wire: **no voltage drop**. Ground symbol: the **reference node** (0 V). Every ground symbol is the **same node**.

Counting check: meshes = $b - n + 1$

Engineering shortcut: count check

For a flat circuit: $\# \text{ meshes} = b - n + 1$

Two slides back: $b = 5$, $n = 4$, so meshes = $5 - 4 + 1 = 2$ ✓

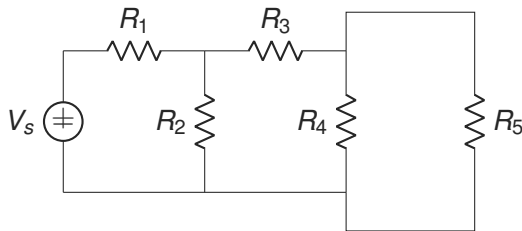
If it doesn't check out, you probably split one node into two.

Week 2 preview: mesh analysis needs 1 equation per mesh, and nodal needs $n - 1$.

Rule

N unknowns need N **independent** equations. Using meshes guarantees that.

Try it: Example 2, count the circuit



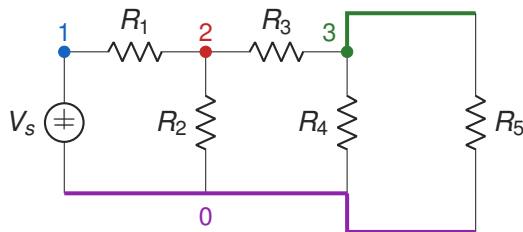
Count:

- (a) nodes
- (b) branches
- (c) meshes

Check with $b - n + 1$.

your work here ↓

Example 2: solution



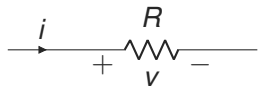
Nodes	4 (1, 2, 3, 0)
Branches	6 (V_s , R_1 to R_5)
Meshes	3
Check	$6 - 4 + 1 = 3 \checkmark$

Trap

The long green wire makes the top of R_5 look like a new node. It isn't: it's node 3. So R_4 and R_5 are in **parallel**.

Resistor, Ohm's law, resistivity

[on formula sheet]



$$v = iR \quad p = vi = i^2 R = \frac{v^2}{R}$$

- $v = iR$ only when the current arrow points **into the + side**. Otherwise $v = -iR$.
- A resistor **always absorbs** power. Negative resistor power means a sign mistake.
- Conductance: $G = 1/R$ (siemens).
- Resistivity: $R = \rho \ell / A$, with ρ = the material's resistivity ($\Omega \cdot \text{m}$; copper $\approx 1.7 \times 10^{-8}$), ℓ = length (m), A = cross-section (m^2). Longer means more R , thicker means less.

Short circuit vs open circuit

wire



Short circuit

$$R = 0$$

voltage across it = 0

any current can flow

gap



Open circuit

$$R = \infty$$

current through it = 0

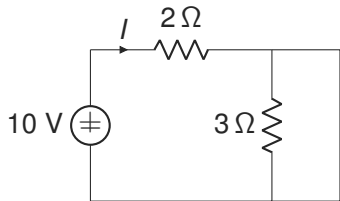
any voltage can appear

Why each one behaves that way

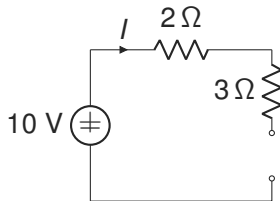
Short: $v = iR = i(0) = 0$ for *any* current, so both ends are one node. **Open:** $i = v/R = v/\infty = 0$ for *any* voltage, so nothing crosses the gap.

Shortcut: short $\Rightarrow$ write “ $v = 0$ ”; open $\Rightarrow$ write “ $i = 0$ ”. Each hands you a free equation.

Try it: Example 3, short and open



(a) wire across the $3\ \Omega$



(b) a gap under the $3\ \Omega$

For each: find I , the voltage across the $3\ \Omega$, and (b) the voltage across the gap.

your work here ↓

Example 3: solution

(a) wire $\parallel 3\ \Omega$: $V_{3\Omega} = 0\ \text{V}$, and no current in the $3\ \Omega$

only the $2\ \Omega$ is left: $I = 10/2 = 5\ \text{A}$

(b) the gap is in series with everything: $I = 0\ \text{A}$

no current means no drops: $V_{3\Omega} = 0\ \text{V}$

KVL: all $10\ \text{V}$ appears across the gap: $10\ \text{V}$

Trap

An open circuit is not “ $0\ \text{V}$ ”. It is “ $0\ \text{A}$ ”. The voltage across a gap can be large.

Ideal sources: what's fixed, what's free



Voltage source

Fixed: the **voltage** across it

Free: its **current** (+, −, or 0)

Killed (set to 0) → **short**



Current source

Fixed: the **current** through it

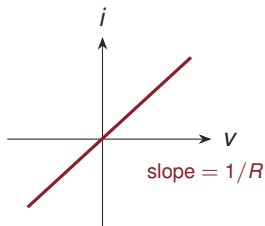
Free: its **voltage** (you solve for it)

Killed (set to 0) → **open**

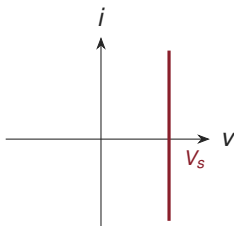
Common exam question

“Find the voltage across the current source”: the source doesn’t tell you. Use KVL around a loop that contains it (Example 4).

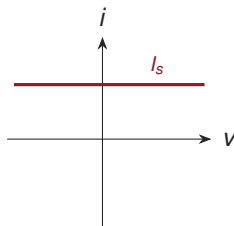
$i-v$ graphs of the three basic elements



Resistor



Voltage source



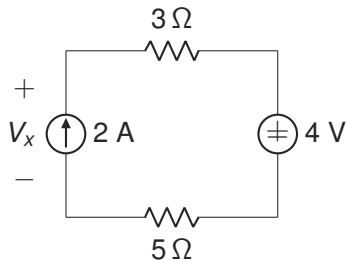
Current source

Voltage source: the voltage is always V_s , whatever the current (vertical line).

Current source: the current is always I_s , whatever the voltage (horizontal line).

Resistor: a straight line through the origin, which is Ohm's law.

Try it: Example 4, voltage across an I source



Find V_x (top is $+$) and the power of the current source. Check the power balance.

your work here ↓

Example 4: solution (1/2)

Current one single loop, so every element carries the source's 2 A (clockwise)

Walk start at the bottom of the source, go up through it (V_x is the rise, $-$ to $+$), then clockwise: drop $3(2)$ across the $3\ \Omega$, drop 4 V entering the 4 V source's $+$, drop $5(2)$ across the $5\ \Omega$

KVL rise = drops: $V_x = 3(2) + 4 + 5(2) = 6 + 4 + 10 \Rightarrow$

$$V_x = 20\text{ V}$$

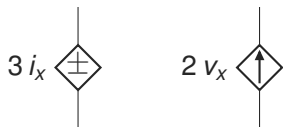
Example 4: solution (2/2)

I source 2 A leaves its + terminal: $p_{abs} = -(20)(2) = -40 \text{ W} \Rightarrow$
supplies 40 W

Absorbed 3Ω : $2^2 \times 3 = 12 \text{ W}$; 5Ω : $2^2 \times 5 = 20 \text{ W}$; 4 V source
(current enters +): $4 \times 2 = 8 \text{ W}$

Check $12 + 20 + 8 = 40 \text{ W} = \text{supplied } \checkmark$

Dependent (controlled) sources



diamond = dependent

	Output	Depends on	Value
VCVS	voltage	voltage v_x	μv_x
CCVS	voltage	current i_x	$r i_x$ (Ω)
VCCS	current	voltage v_x	$g v_x$ (S)
CCCS	current	current i_x	βi_x

The two rules

1. In KCL/KVL, treat it like a normal source whose value is an expression. Add one equation that links i_x or v_x to your unknowns.
2. It is **never killed** when finding R_{eq} or R_{Th} (week 3). Only independent sources get killed.

Dependent sources, in words

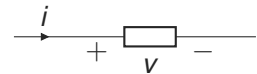
A dependent source's value is set by a voltage or current **somewhere else** in the circuit (the **control**). The control is always labeled on the drawing (i_x through some element, or v_x across it).

Name	Meaning of the letters	Gain	Unit of gain
VCVS	Voltage-Controlled Voltage Source	μ (voltage gain)	none
CCVS	Current-Controlled Voltage Source	r (transresistance)	Ω (V per A)
VCCS	Voltage-Controlled Current Source	g (transconductance)	S (A per V)
CCCS	Current-Controlled Current Source	β (current gain)	none

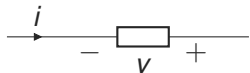
Example: a CCVS labeled $3i_x$ makes a voltage of 3 V for every 1 A of i_x . If $i_x = 2$ A, it's a 6 V source at that moment. They model transistors and amplifiers.

Power: the passive sign convention

[NOT on formula sheet: memorize]



i enters $+$ $\Rightarrow p = +vi$



i enters $-$ $\Rightarrow p = -vi$

p = power **absorbed**

$p > 0 \Rightarrow$ **absorbs**

$p < 0 \Rightarrow$ **supplies**

Engineering shortcut

Always use the rule, then let the sign say absorb or supply. Never guess from “it’s a source”.

Why the sign rule works

Current flowing **into the + terminal** means charge moves from high voltage to low voltage **through** the element. Each coulomb loses v joules there, so the element **takes** energy: $p = +vi$ absorbed. That's what a resistor always does, which is why it's called the **passive** convention.

Current flowing **out of the + terminal** means charge is pushed **up** in voltage by the element: the element **gives** energy (a battery driving a circuit). Then $p = -vi$ absorbed, which is negative, meaning it supplies.

One-line summary

Pick the formula from the arrow and the $+$ sign, plug in the numbers **with their signs**, and read the result: positive absorbs, negative supplies.

Power balance, and sources that absorb

Conservation of energy

$$\sum p_{\text{absorbed}} = 0 \quad \Longleftrightarrow \quad \sum p_{\text{supplied}} = \sum p_{\text{absorbed}}$$

Free answer check

On any “find the power” question, add everything up. If it doesn’t balance, something is wrong. It takes 20 seconds.

Sources don’t always supply

A source can **absorb** power (a battery being charged). Exams often ask you to “identify the source(s) of power” to test exactly this.

Try it: Example 5, passive sign convention

Find the power absorbed. Absorbing or supplying?

(a) $v = 10 \text{ V}$, $i = 2 \text{ A}$ into $+$ (b) $v = 10 \text{ V}$, $i = 2 \text{ A}$ into $-$

(c) $v = -5 \text{ V}$, $i = 3 \text{ A}$ into $+$ (d) $v = -5 \text{ V}$, $i = -3 \text{ A}$ into $+$

your work here ↓

Example 5: solution

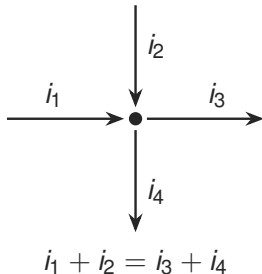
Current enters	Formula	Result
(a) +	$p = vi = (10)(2) = +20 \text{ W}$	absorbs 20 W
(b) -	$p = -vi = -(10)(2) = -20 \text{ W}$	supplies 20 W
(c) +	$p = vi = (-5)(3) = -15 \text{ W}$	supplies 15 W
(d) +	$p = vi = (-5)(-3) = +15 \text{ W}$	absorbs 15 W

Why (c) works

A negative voltage means the real + is on the other terminal. Don't redraw anything. Plug in the signs and the algebra handles it.

KCL: current in = current out

[on formula sheet]



$$\sum i_{\text{in}} = \sum i_{\text{out}}$$

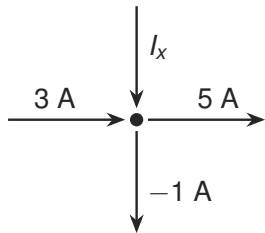
at **every node** (conservation of charge).

Sheet form: $\sum i_n = 0$ (in +, out -).

Engineering shortcut: KCL on a whole region

KCL works on any closed boundary around part of a circuit (a **supernode**):
current in = current out, without solving inside.

Try it: Example 6, KCL



(a)

Find I_x and I_y .



(b) unknown circuit inside

your work here ↓

Example 6: solution

(a) in = out: $3 + I_x = 5 + (-1) = 4 \Rightarrow I_x = 1 \text{ A (into the node)}$

(b) KCL on the boundary: $2 + 7 = 4 + I_y \Rightarrow I_y = 5 \text{ A}$

Trap

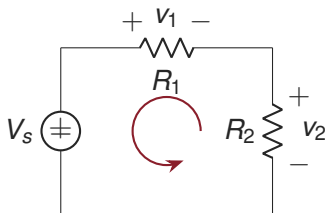
“−1 A out” is really 1 A *in*. Keep the sign as given and let the algebra do it.

Engineering shortcut

In (b) you never needed to know what was inside the box.

KVL: around any loop, voltages add to zero

[on formula sheet]



Walk the loop in one direction:

enter $-$, exit $+$ $\Rightarrow$ **rise**: write $+v$

enter $+$, exit $-$ $\Rightarrow$ **drop**: write $-v$

Clockwise here: $+V_s - v_1 - v_2 = 0$

Engineering shortcut

Across a resistor: walking *with* your current $= -iR$, walking *against* it $= +iR$.

Why KCL and KVL are always true

KCL: charge can't pile up

A node is just a wire junction. It can't store charge, so every coulomb that arrives each second must leave in that same second. That's "current in = current out".

KVL: energy is conserved

Voltage is energy per charge. Follow one coulomb around a closed loop back to its starting point: it ends where it began, at the same voltage. So all the rises and drops along the way must cancel: $\sum v = 0$.

Both laws hold for any circuit, whatever the elements are (resistors, sources, capacitors, anything). That's why every method in this course is built on them.

Assumed directions and negative answers

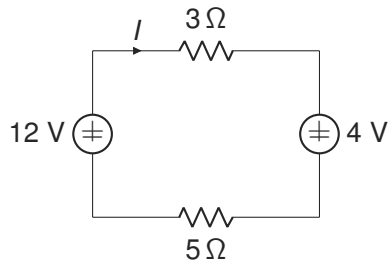
You pick the direction. Being wrong is fine.

Before solving, draw an arrow for every unknown current, any direction you like. If the answer comes out negative, the real current flows the other way with that size.

Common exam wording: “identify the direction of flow”

Don't just write $I = -2 \text{ A}$. Write: $I = -2 \text{ A}$, so 2 A flows right to left .

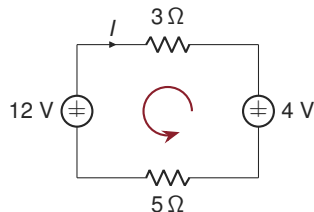
Try it: Example 7, one loop, two sources



Find I and the power of every element. Check the power balance.

your work here ↓

Example 7: solution



Walk (cw) up the 12 V (– to +): +12; through $3\ \Omega$ with I : $-3I$; into the 4 V's +: -4 ; through $5\ \Omega$: $-5I$

KVL $+12 - 3I - 4 - 5I = 0$

Solve $8I = 8 \Rightarrow I = 1\text{ A clockwise}$

12 V I leaves +: **supplies 12 W**

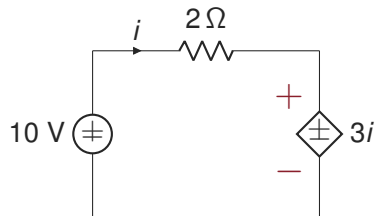
4 V I enters +: **absorbs 4 W**

Resistors $3(1)^2 = 3\text{ W}$, $5(1)^2 = 5\text{ W}$

Check $12 = 4 + 3 + 5\ \checkmark$

If you assumed counterclockwise, you'd get $I = -1\text{ A}$: the same physical answer.

Try it: Example 8, dependent source in a loop



The $+$ of the $3i$ source is on top.

Find i and the power of each element.

your work here ↓

Example 8: solution

Walk (cw) up the 10 V: $+10$; through 2Ω with i : $-2i$; down through the $3i$ source entering its $+$: $-3i$

KVL $+10 - 2i - 3i = 0 \Rightarrow 5i = 10 \Rightarrow i = 2\text{ A}$

Value the dependent source is now $3i = 3(2) = 6\text{ V}$

10 V i leaves its $+$: $p_{abs} = -(10)(2) = -20\text{ W}$, it **supplies 20 W**

2Ω $i^2 R = 2^2 \times 2 = 8\text{ W}$ absorbed

$3i$ source i enters its $+$: $p_{abs} = +(6)(2) = 12\text{ W}$ absorbed

Check $20 = 8 + 12 \checkmark$

The control i was already the unknown. If the control were some v_x elsewhere, add one more equation linking v_x to the unknowns (like $v_x = R \times$ its current).

Series vs parallel: the definitions that work

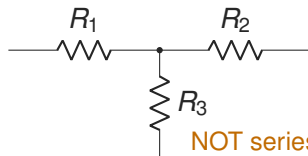
The definitions

Series: share **one node**, and **nothing else** touches that node $\Rightarrow$ same current.

Parallel: connected between the **same two nodes** $\Rightarrow$ same voltage.

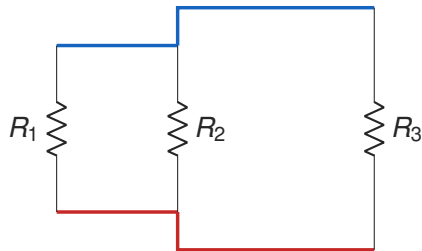


series ✓



R_3 touches the middle node, so the current can split there. This is the #1 R_{eq} mistake.

Spotting parallel: color the nodes



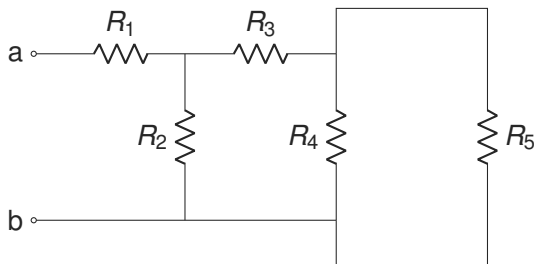
Every resistor touches **blue** and **red**.

Only 2 nodes, so all three are **in parallel**, even though R_3 is drawn far away.

Engineering shortcut: highlighter method

On the exam, trace each node in a different color (or number them). Two resistors with the *same pair of colors* are in parallel. The shape of the drawing means nothing, only the colors do.

Try it: Example 9, spot series and parallel

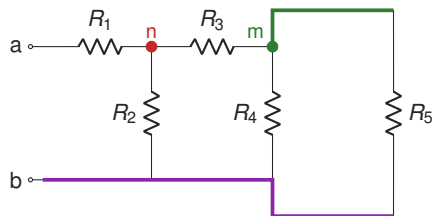


For each pair, say series, parallel, or neither:

- (a) R_4 and R_5
- (b) R_3 and R_4
- (c) R_1 and R_2
- (d) R_1 and R_3

your work here ↓

Example 9: solution



- (a) parallel both go from m to b
(b) neither R_5 also touches m
(c) neither R_3 also touches n
(d) neither R_2 also touches n

After combining $R_4 \parallel R_5$, the result **is** in series with R_3 . That's why you redraw after every step.

Combining resistors

[on formula sheet]

$$\text{Series: } R_{eq} = R_1 + R_2 + \dots \quad \text{Parallel: } \frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$$

Parallel shortcuts (memorize)

$$\text{Two resistors } R_1 \parallel R_2 = \frac{R_1 R_2}{R_1 + R_2} \quad (\text{only for TWO})$$

$$N \text{ equal } R \quad R/N: \quad 10 \parallel 10 = 5, \quad 6 \parallel 6 \parallel 6 = 2$$

$$R \parallel 2R = \frac{2}{3}R: \quad 1 \text{ k} \parallel 2 \text{ k} = 0.667 \text{ k}$$

$$R \parallel 0 = 0 \quad (\text{a short kills everything across it})$$

Sanity check: parallel $<$ smallest R , series $>$ largest R .

Where the series rule comes from

Series: same current

Both resistors carry the same i . KVL around the loop:

$$v = iR_1 + iR_2 = i(R_1 + R_2)$$

Compared with $v = iR_{eq}$: the pair acts exactly like one resistor $R_{eq} = R_1 + R_2$.

Where the parallel rule comes from

Parallel: same voltage

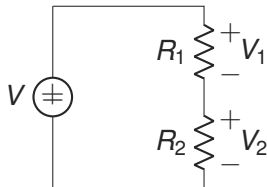
Both resistors see the same v . KCL at the top node:

$$i = \frac{v}{R_1} + \frac{v}{R_2} = v \left(\frac{1}{R_1} + \frac{1}{R_2} \right)$$

Compared with $i = v/R_{eq}$: $\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$. The **conductances** ($1/R$) add.

Voltage divider (series)

[on formula sheet]



$$V_2 = V \frac{R_2}{R_1 + R_2}$$

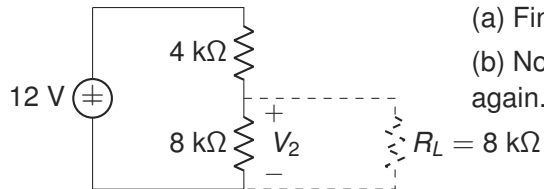
Same resistor on top. Any number of resistors: $V_m = V \frac{R_m}{R_{eq}}$

Engineering shortcut

Bigger R gets bigger V . Equal resistors split it in half. R and $2R$ split it 1/3 and 2/3.

Trap: only valid if nothing else connects to the middle node (Example 10).

Try it: Example 10, the loaded divider trap



(a) Find V_2 **without** R_L .

(b) Now connect R_L (dashed). Find V_2 again.

your work here ↓

Example 10: solution

(a) divider: $V_2 = 12 \cdot \frac{8}{4 + 8} = 8 \text{ V}$

(b) R_L is in parallel with the 8 k: $8 \parallel 8 = 4 \text{ k}$

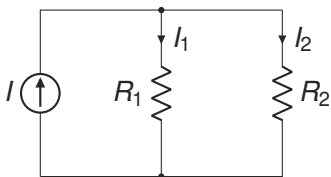
now divide against that: $V_2 = 12 \cdot \frac{4}{4 + 4} = 6 \text{ V}$

Trap

Adding a load **changes** the divider. Combine the load with the resistor it sits across first, then divide.

Current divider (parallel)

[on formula sheet]



$$I_1 = I \frac{R_2}{R_1 + R_2}$$

Other resistor on top (two only).

$$\text{Any number: } I_m = I \frac{G_m}{G_{eq}}$$

The flip

Smaller R gets bigger I , the opposite of the voltage divider.

Engineering shortcut

Often faster: $V = I \cdot R_{eq}$ first, then V/R per branch.

Where the voltage divider comes from

Voltage divider: the same resistor on top

One current flows through the whole chain, so

$$i = \frac{V}{R_1 + R_2} \quad \Rightarrow \quad V_2 = iR_2 = V \frac{R_2}{R_1 + R_2}$$

Each resistor takes the share of V that its resistance is of the total.

Where the current divider comes from

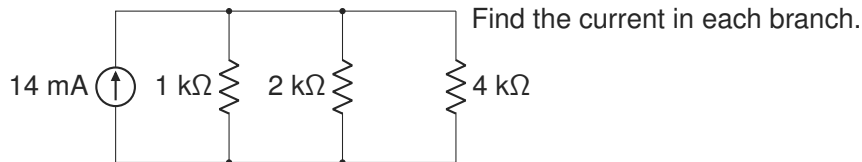
Current divider: the *other* resistor on top

Both branches share one voltage, $V = I \frac{R_1 R_2}{R_1 + R_2}$, so

$$I_1 = \frac{V}{R_1} = I \frac{R_1 R_2}{R_1(R_1 + R_2)} = I \frac{R_2}{R_1 + R_2}$$

The R_1 cancels, and the other resistor is left on top.

Try it: Example 11, current divider



your work here ↓

Example 11: solution

Conductances $G = 1 + 0.5 + 0.25 = 1.75 \text{ mS}$

Shared voltage $V = I/G_{eq} = 14/1.75 = 8 \text{ V}$

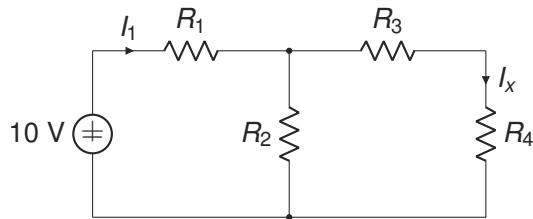
Each branch $8/1 = 8 \text{ mA}$, $8/2 = 4 \text{ mA}$, $8/4 = 2 \text{ mA}$

Check (KCL) $8 + 4 + 2 = 14 \text{ mA}$ ✓

Engineering shortcut

Voltage first, then Ohm's law per branch. There's no divider formula to flip by mistake, and it works for any number of branches.

Try it: Example 12, ladder circuit



All $R = 2\text{ k}\Omega$.

Find I_x .

your work here ↓

Example 12: solution (1/2)

Plan collapse the ladder into one resistor to get I_1 , then split currents back out

1. Series $R_3 + R_4 = 2 + 2 = 4$ k: nothing else touches the node between them

2. Parallel that 4 k and R_2 sit between the same two nodes:
$$\frac{2 \cdot 4}{2 + 4} = \frac{4}{3} \text{ k}$$

3. Series add R_1 : total $= 2 + \frac{4}{3} = \frac{10}{3}$ k

Example 12: solution (2/2)

4. Source / Ohm: $I_1 = 10 \text{ V} / \frac{10}{3} \text{ k} = 3 \text{ mA}$

5. Divider I_1 splits between R_2 (2 k) and the $R_3 + R_4$ branch (4 k);
other resistor on top: $I_x = 3 \cdot \frac{2}{2 + 4} = I_x = 1 \text{ mA}$

Check both branches have the same voltage: $(3 - 1)(2) = 4 \text{ V}$
and $(1)(4) = 4 \text{ V} \checkmark$

Ladder shortcut: assume $I_x = 1$

Engineering shortcut: unit output method

Circuits are linear: **pretend** $I_x = 1$, walk back to the source, see what V_s you'd need, then scale.

Assume $I_x = 1 \text{ mA} \Rightarrow V_{R_3+R_4} = (1)(4) = 4 \text{ V}$

R_2 same 4 V (parallel) $\Rightarrow I_2 = 4/2 = 2 \text{ mA}$

KCL $I_1 = 1 + 2 = 3 \text{ mA} \Rightarrow V_{R_1} = 6 \text{ V}$

KVL needed $V_s = 6 + 4 = 10 \text{ V}$

Scale real V_s is also 10 V, so factor = 1 $\Rightarrow I_x = 1 \text{ mA}$

(Source of 20 V instead? Factor $20/10 = 2$, so $I_x = 2 \text{ mA}$.)

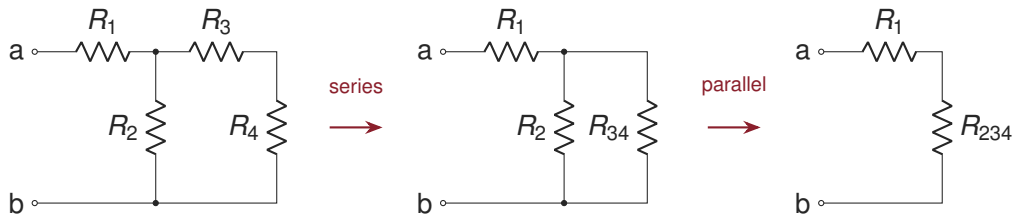
Equivalent resistance: the method

- 1 Color / label every node.
- 2 Start at the side **farthest** from the terminals a–b.
- 3 Combine one clear series or parallel pair.
- 4 **Redraw** the smaller circuit. Repeat.
- 5 Sanity check: parallel $<$ smallest, series $>$ largest.

Engineering shortcut

Redraw after every combine. It feels slow, but it's where the marks are and it stops double-counting mistakes.

The method in pictures

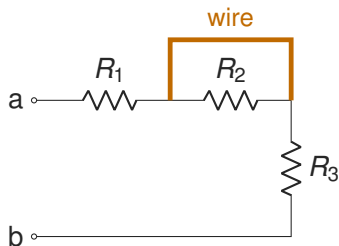


$$R_{34} = R_3 + R_4, \quad R_{234} = R_2 \parallel R_{34}, \quad \text{then } R_{ab} = R_1 + R_{234}.$$

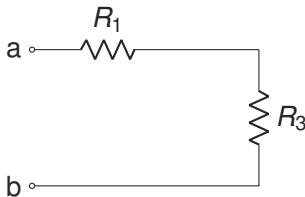
Notice each drawing is simpler than the last. Never skip a drawing on the exam.

R_{eq} trap 1: a shorted resistor disappears

Drawn like this



Is really this

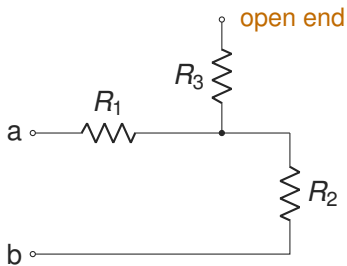


The wire ties both ends of R_2 to the same node, so $V_{R_2} = 0$. By Ohm's law its current is $0/R_2 = 0$: it carries nothing, and every bit of current takes the wire instead. Delete R_2 and redraw: $R_{ab} = R_1 + R_3$.

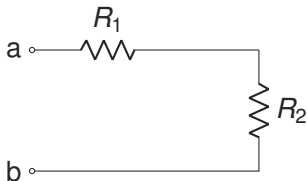
The same argument works for a closed switch, or for any two nodes joined by a wire.

R_{eq} trap 2: a dangling resistor

Drawn like this

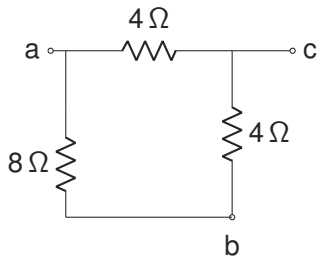


Is really this



One end of R_3 goes nowhere, so no current can flow through it. Delete it. It does **not** add to the chain: $R_{ab} = R_1 + R_2$.

R_{eq} trap 3: the terminals matter



Same 3 resistors, 3 different questions:

$$R_{ab} \quad 8 \parallel (4 + 4) = 4 \, \Omega$$

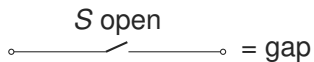
$$R_{ac} \quad 4 \parallel (8 + 4) = 3 \, \Omega$$

$$R_{cb} \quad 4 \parallel (4 + 8) = 3 \, \Omega$$

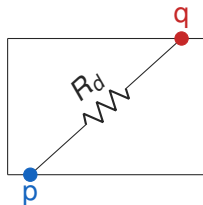
Engineering shortcut

Put your finger on the two terminals. The resistor directly between them is in parallel with “everything else” between them.

R_{eq} trap 4: switches and diagonals



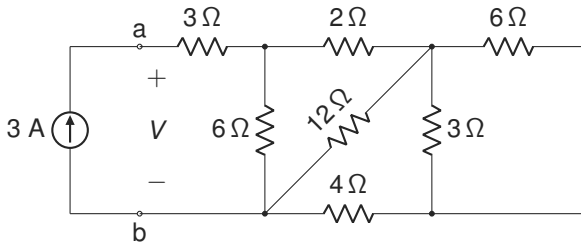
Switches: redraw the circuit once for **each** switch case. Exams love this.



Diagonal resistors are nothing special. Find the two nodes they connect (here p and q) and treat it like any other resistor.

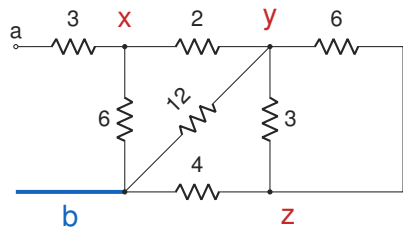
Try it: Example 13, exam-style R_{eq}

(a) Find R_{eq} to the right of a–b. (b) Use it to find V .



your work here ↓

Example 13: solution (1/2)



y-z $3 \parallel 6 = 2$

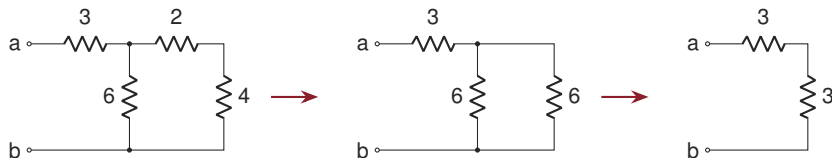
y-b $2 + 4 = 6$ (only 2 things touch z)

y-b diagonal $12 \parallel 6 = 4$

Engineering shortcut

The whole blue bottom-left wire is node b, including the bottom of the 6 Ω and the low end of the diagonal 12 Ω . Seeing that is the whole question.

Example 13: solution (2/2)

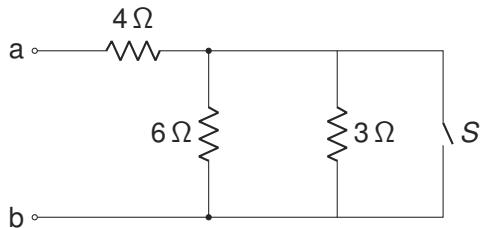


x-b $2 + 4 = 6$, then $6 \parallel 6 = 3$

a-b $3 + 3 = R_{eq} = 6 \Omega$

(b) $V = (3 \text{ A})(6 \Omega) = V = 18 \text{ V}$

Try it: Example 14, switch



Find R_{ab} :

- (a) S open
- (b) S closed

your work here ↓

Example 14: solution (1/2)

(a) open an open switch is a gap: it carries nothing, so delete it

$6\ \Omega \parallel 3\ \Omega = \frac{18}{9} = 2\ \Omega$, then in series with $4\ \Omega$:

$$R_{ab} = 6\ \Omega$$

Example 14: solution (2/2)

(b) closed a closed switch is a wire. It connects the same two nodes as the $6\ \Omega$ and $3\ \Omega$

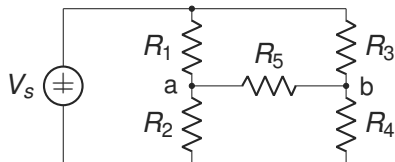
anything in parallel with a wire has $0\ \text{V}$ across it and carries no current, so $6 \parallel 3 \parallel 0 = 0$

only the $4\ \Omega$ is left: $R_{ab} = 4\ \Omega$

Trap

A closed switch shorts **everything** in parallel with it, not just the resistor drawn next to it.

Wheatstone bridge



R_5 connects the middle of both sides.
It's **not** series or parallel with anything, so the usual combining fails.

Balanced when

$$\frac{R_1}{R_2} = \frac{R_3}{R_4}$$

Then $v_a = v_b$, so **no current** flows in R_5 .

Engineering shortcut: balanced bridge

Check the ratio first. If it's balanced, delete R_5 (or short it, same answer).
What's left is simple series/parallel.

Why the balance condition works

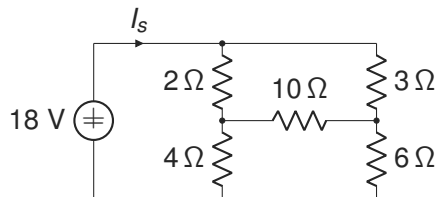
Suppose R_5 carries no current. Then each side is just a two-resistor divider from V_s to ground:

$$v_a = V_s \frac{R_2}{R_1 + R_2}, \quad v_b = V_s \frac{R_4}{R_3 + R_4}$$

These are equal exactly when $\frac{R_1}{R_2} = \frac{R_3}{R_4}$. With $v_a = v_b$, the middle resistor has 0 V across it and so really does carry no current: the assumption checks out.

That's why a balanced bridge ignores R_5 , and why bridges **measure** so well: adjust a known resistor until a meter in the middle reads zero, then the ratio equation gives the unknown exactly.

Try it: Example 15, bridge



Find the current in the $10\ \Omega$, the equivalent resistance seen by the source, and I_s .

your work here ↓

Example 15: solution

Ratio $2/4 = 0.5$ and $3/6 = 0.5 \Rightarrow$ balanced

$10\ \Omega$ $I_{10} = 0$, so delete it

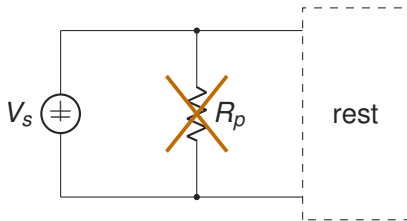
Left side $2 + 4 = 6\ \Omega$ **Right side:** $3 + 6 = 9\ \Omega$

R_{eq} $6 \parallel 9 = 54/15 = 3.6\ \Omega$

I_s $18/3.6 = 5\text{ A}$

Check $V_a = 18 \times 4/6 = 12\text{ V}$, $V_b = 18 \times 6/9 = 12\text{ V}$, equal ✓

Cross-out trap 1: R across a V source

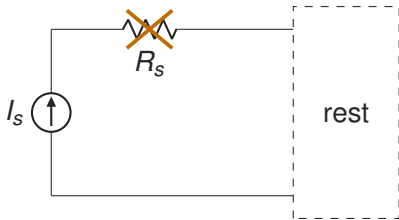


R_p always has exactly V_s across it, no matter what the rest does.

So for the **rest of the circuit**, R_p changes nothing. Cross it out.

It only changes the **source's own current**: $I_{\text{source}} = I_{\text{rest}} + V_s/R_p$.

Cross-out trap 2: R in series with an I source

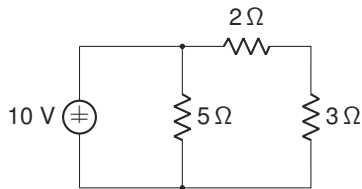


R_s always carries exactly I_s , no matter what the rest does.

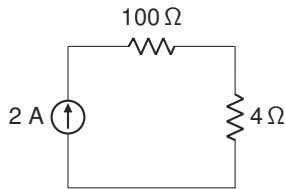
So for the **rest of the circuit**, R_s acts like a wire. Cross it out.

It only changes the **source's own voltage**: $V_{\text{source}} = V_{\text{rest}} + I_s R_s$.

Try it: Example 16, cross out



(a)



(b)

(a) Current in the $3\ \Omega$, and the source current.
and across the current source.

(b) Voltage across the $4\ \Omega$,

your work here ↓

Example 16: solution

(a) $5\ \Omega$ is across the $10\ \text{V}$ source: cross it out for the rest

$$I_{3\Omega} = 10/(2 + 3) = 2\ \text{A}$$

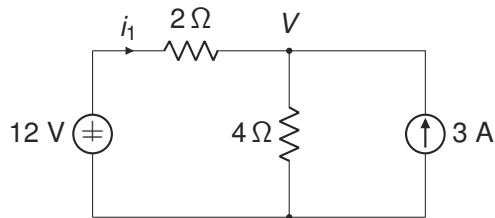
$5\ \Omega$ draws $10/5 = 2\ \text{A}$, so the source gives $2 + 2 = 4\ \text{A}$

(b) $100\ \Omega$ is in series with the $2\ \text{A}$ source: cross it out for the rest

$$V_{4\Omega} = 2 \times 4 = 8\ \text{V} \quad (\text{the } 100\ \Omega \text{ changes nothing here})$$

$$\text{source voltage} = 2 \times (100 + 4) = 208\ \text{V} \quad (\text{top } +)$$

Try it: Example 17, both kinds of source



Bottom = 0 V.

Find V and the power of each source.

your work here ↓

Example 17: solution (1/2)

Plan one unknown node voltage V (bottom = 0 V): one KCL equation at that node

Currents leaving through the $4\ \Omega$: $V/4$. Entering from the 12 V side through the $2\ \Omega$: $(12 - V)/2$. Entering from the 3 A source: 3 A

KCL out = in: $\frac{V}{4} = \frac{12 - V}{2} + 3$

Solve $\times 4$: $V = 24 - 2V + 12 \Rightarrow 3V = 36 \Rightarrow V = 12\text{ V}$

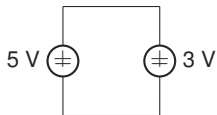
Example 17: solution (2/2)

i_1 $(12 - 12)/2 = 0$: the 12 V source carries no current, so it gives **zero** power

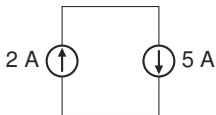
3 A source it pushes 3 A out of its + at 12 V: **supplies 36 W**

Check the $4\ \Omega$ absorbs $12^2/4 = 36\text{ W}$ ✓

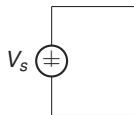
Impossible circuits



different V sources in
parallel: **breaks KVL**



different I sources in
series: **breaks KCL**



wire across a V source:
infinite current

If you think you see one of these on an exam, you misread the circuit. (A wire across a *current* source is fine: all the current takes the wire.)

Sign and formula traps

Check these before you box an answer

- $v = iR$ needs i entering the $+$ side. Otherwise $v = -iR$.
- Product-over-sum is for **two** resistors only.
- The current divider uses the **other** R on top. The voltage divider uses the **same** R .
- Resistor power is always ≥ 0 . Source power can be either sign.
- Every ground symbol is the same node.
- A negative answer is fine: state the real direction.

Exam-format traps

Using a different method than the one requested can get 0 marks

If they say nodal, use nodal, even if something else is faster.

“First give the equations with variables, then replace with values”

Write the symbolic equation first. It's worth marks by itself.

Box your final answer

Include units, and a direction for currents.

Summary (1/2)

Basics

$$i = dq/dt, \quad p = vi, \quad W = Pt, \quad 1 \text{ A}\cdot\text{h} = 3600 \text{ C}, \quad R = \rho\ell/A$$

Ohm

$$v = iR \text{ (} i \text{ into +)}, \quad p = vi = i^2 R = v^2/R$$

PSC

$$i \text{ into +: } p = vi. \quad p > 0 \text{ absorbs, } p < 0 \text{ supplies. } \sum p = 0$$

KCL / KVL

$$\sum i_{in} = \sum i_{out} \text{ (also on regions)} \quad / \quad \sum v = 0 \text{ (rise +, drop -)}$$

Counting

$$\text{meshes} = b - n + 1 \quad \bullet \quad \text{one wire} = \text{one node}$$

Series

$$\text{one node, nothing else attached, same } i, \quad R_{eq} = \sum R$$

Parallel

$$\text{same two nodes, same } v, \text{ two: } \frac{R_1 R_2}{R_1 + R_2}$$

Summary (2/2)

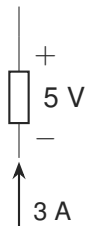
Dividers	$V_m = V \frac{R_m}{R_{eq}}$ (same R) • $I_1 = I \frac{R_2}{R_1 + R_2}$ (other R) • a load changes the divider
Sources	V fixes v (its i is free), I fixes i (its v is free). Killed: $V \rightarrow$ short, $I \rightarrow$ open. Dependent: never killed
R_{eq}	color nodes, far end first, redraw each step. Delete shorted and dangling R . Terminals matter. Balanced bridge: delete the middle
Cross out	R across a V source, and R in series with an I source
Shortcuts	$k\Omega \cdot \text{mA} = V$ • R/N • voltage first, then V/R • ladder: assume $I_x = 1$
Checks	KCL, KVL, power balance, parallel $<$ smallest, series $>$ largest

Final check

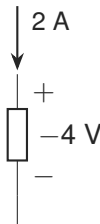
8 problems. One per slide, with space. No looking back.

If one takes more than 10 min, go back to that section. The answer sheet comes after P8.

P1: sign convention



(a)

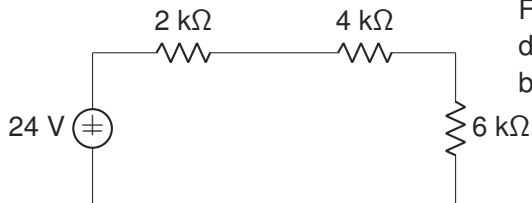


(b)

Power absorbed by each? Absorbing or supplying?

your work here ↓

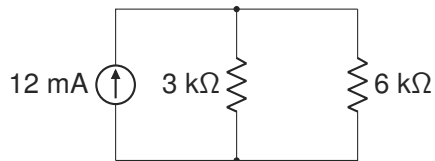
P2: series circuit



Find the current, V_{4k} (use the divider), and the power supplied by the source.

your work here ↓

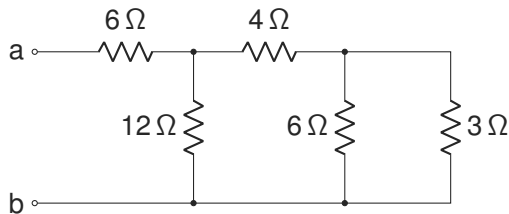
P3: current divider



Find each branch current and the voltage across them.

your work here ↓

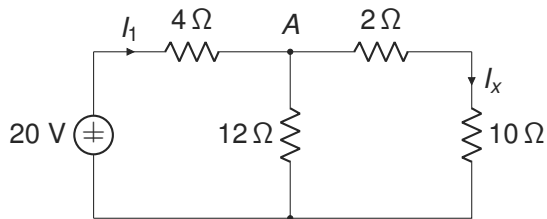
P4: equivalent resistance



Find R_{ab} .

your work here ↓

P5: ladder

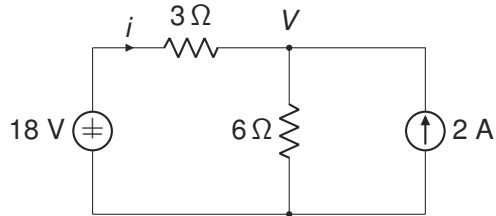


Find I_1 , V_A , I_x .

(Try the “assume $I_x = 1$ ”
trick.)

your work here ↓

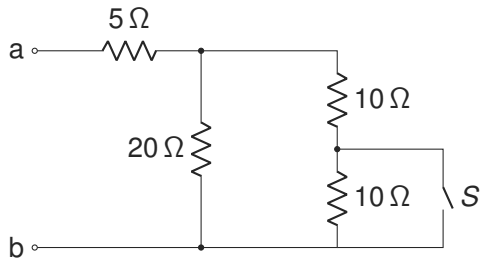
P6: two sources



Find V and the power supplied by each source. Check the balance.

your work here ↓

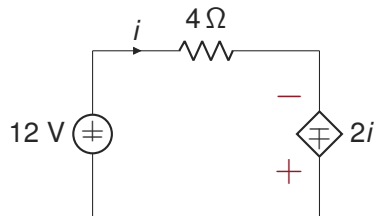
P7: switch



Find R_{ab} with S open and with S closed.

your work here ↓

P8: dependent source that helps



The $+$ of the $2i$ source is at the **bottom**.

Find i . Does the dependent source absorb or supply, and how much?

your work here ↓

Answer sheet

Full worked solutions to P1–P8.

P1: solution

(a) direction the arrow enters the **bottom** terminal, which is marked $-$, so the current **leaves** the $+$ terminal

(a) formula leaving $+$ means the PSC needs a minus: $p_{abs} = -vi$

(a) numbers $p_{abs} = -(5)(3) = -15 \text{ W}$

(a) read it negative absorbed power means it is delivering:

supplies 15 W

(b) direction the arrow enters the $+$ terminal, so $p_{abs} = +vi$ directly

(b) numbers $p_{abs} = (-4)(2) = -8 \text{ W}$ (the voltage itself is negative here)

(b) read it negative again: supplies 8 W

P2: solution

Total R $2 + 4 + 6 = 12 \text{ k}\Omega$

Current $I = 24/12 = 2 \text{ mA}$ ($\text{V/k}\Omega = \text{mA}$)

Divider $V_{4k} = 24 \cdot \frac{4}{12} = 8 \text{ V}$ (check: $2 \text{ mA} \times 4 \text{ k} = 8 \text{ V}$ ✓)

Power $P = 24 \times 2 \text{ mA} = 48 \text{ mW supplied}$

Check $I^2 R: 8 + 16 + 24 = 48 \text{ mW}$ ✓

P3: solution

Plan both resistors share the same voltage (parallel), so find that voltage first, then use Ohm's law on each branch

R_{eq} product over sum: $3 \parallel 6 = \frac{3 \times 6}{3 + 6} = \frac{18}{9} = 2 \text{ k}\Omega$

Voltage the source pushes all 12 mA through that equivalent:

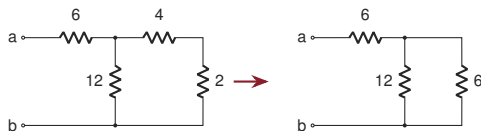
$$V = IR_{eq} = 12 \text{ mA} \times 2 \text{ k}\Omega = 24 \text{ V}$$

Branch 1 $I_{3k} = V/R = 24/3 = 8 \text{ mA}$

Branch 2 $I_{6k} = 24/6 = 4 \text{ mA}$

Check KCL: $8 + 4 = 12 \text{ mA}$ ✓ and the smaller resistor took the bigger share ✓

P4: solution (1/2)



Plan work from the end **farthest** from a–b and redraw after each combine

Step 1 right-hand 6 Ω and 3 Ω are between the same two nodes:
 $6 \parallel 3 = 2 \Omega$

Step 2 that 2 Ω shares one node with the 4 Ω, nothing else attached:
 $4 + 2 = 6 \Omega$

P4: solution (2/2)

Step 3 now between the same nodes as the $12\ \Omega$: $12 \parallel 6 = 4\ \Omega$

Step 4 in series with the input $6\ \Omega$: $6 + 4 = R_{ab} = 10\ \Omega$

P5: solution (two ways)

Collapse

1 $2 + 10 = 12$

2 $12 \parallel 12 = 6$

3 $4 + 6 = 10 \Omega$

4 $I_1 = 20/10 = 2 \text{ A}$

5 $V_A = 20 - 4 \cdot 2 = 12 \text{ V}$

6 $I_x = 12/12 = 1 \text{ A}$

Assume $I_x = 1 \text{ A}$

1 $V_A = 1 \cdot (2 + 10) = 12 \text{ V}$

2 $I_{12} = 12/12 = 1 \text{ A}$

3 $I_1 = 1 + 1 = 2 \text{ A}$

4 $V_s = 12 + 4 \cdot 2 = 20 \text{ V}$

5 matches, so factor = 1 ✓

P6: solution (1/2)

Plan one unknown node voltage V : KCL at that node, currents out = currents in

KCL out through $6\ \Omega$: $V/6$; in through $3\ \Omega$: $(18 - V)/3$; in from the source: $2\text{ A} \Rightarrow \frac{V}{6} = \frac{18 - V}{3} + 2$

Solve $\times 6$: $V = 36 - 2V + 12 \Rightarrow 3V = 48 \Rightarrow V = 16\text{ V}$

i $(18 - 16)/3 = 2/3\text{ A}$, flowing out of the 18 V source's +

P6: solution (2/2)

18 V current leaves $+$: supplies $18 \times \frac{2}{3} =$ **12 W**

2 A pushes 2 A out of its top at 16 V: supplies $2 \times 16 =$
32 W

Absorbed $6\ \Omega$: $16^2/6 = 42.67\text{ W}$; $3\ \Omega$: $(\frac{2}{3})^2 \times 3 = 1.33\text{ W}$

Check $12 + 32 = 44 = 42.67 + 1.33\ \checkmark$

P7: solution

Read it the $20\ \Omega$ and the right branch (two $10\ \Omega$ in series) are in parallel; S is across the lower $10\ \Omega$ only

Open S is a gap: right branch $= 10 + 10 = 20$, $20 \parallel 20 = 10$,
 $R_{ab} = 5 + 10 = 15\ \Omega$

Closed S is a wire across the lower $10\ \Omega$, which disappears: right branch $= 10$

$$20 \parallel 10 = \frac{200}{30} = 6.67, \quad R_{ab} = 5 + 6.67 = 11.67\ \Omega$$

Trap

Here the switch is across *one* resistor, so it removes only that one. In Example 14 it was across the whole parallel group.

P8: solution

KVL (cw) up the 12 V: $+12$, $4\ \Omega$: $-4i$, down through $2i$ entering its $-$: $+2i$

Equation $12 - 4i + 2i = 0 \Rightarrow i = 6\text{ A}$

Dep. source i enters its $-$: $p = -(2i)(i) = -72\text{ W} \Rightarrow$
supplies 72 W

Check 12 V supplies 72 W, resistor absorbs $4 \cdot 36 = 144\text{ W}$:
 $72 + 72 = 144\ \checkmark$

Dependent sources can supply power too. Only the sign decides.